# Lili Quan


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Singapore Management University Research Scientist 1997-03-12
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## Education/Research Experience

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| now<br>2026.01 | **Research Scientist**, **Singapore Management University**<br>Group: Prof.Xiaofei Xie |
| 2025.11<br>2021.09 | **PhD.**, Tianjin University<br>School of Computer Science and Technology. Advisor: Prof. Xiaohong Li |
| 2021.09<br>2018.06 | **Master**, Tianjin University<br>School of Computer Science and Technology |
| 2018.09<br>2014.06 | **Bachelor**, Tianjin University of Finance and Economics<br>School of Computer Science and Technology |

## Research Interests

- **LLM Supply Chain Security Analysis**
  - Software Composition Analysis for LLM Applications
  - Vulnerability Discovery and Exploitation in LLM Supply Chains
- **Testing of Intelligent Software Systems**
  - Testing of Deep Learning Frameworks
  - Testing of Deep Learning Models
- **Protocol Security and Testing**
  - Fuzzing for Protocol Software
  - Inconsistency Detection for Protocol Software

## Publications

1. **Lili Quan**, Qianyu Guo, Hongxu Chen, Xiaofei Xie, Xiaohong Li, Yang Liu, and Jing Hu. SADT: syntax-aware differential testing of certificate validation in SSL/TLS implementations. In 2020 35th IEEE/ACM International Conference on Automated Software Engineering (ASE'2020), pages 524-535. IEEE, 2020. [**CCF-A**]
2. **Lili Quan**, Qianyu Guo, Xiaofei Xie, Sen Chen, Xiaohong Li, and Yang Liu. Towards Understanding the Faults of JavaScript-Based Deep Learning Systems. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (ASE '22). [**CCF-A**]
3. **Lili Quan**, Xiaofei Xie, Qianyu Guo, Lingxiao Jiang, Sen Chen, Junjie Wang, and Xiaohong Li. TensorJSFuzz: Effective Testing of Web-Based Deep Learning Frameworks via Input-Constraint Extraction. In Proceedings of the Web Conference, Accepted. (WWW'25). [**CCF-A**]
4. **Lili Quan**, Tianlin Li, Xiaofei Xie, Zhenpeng Chen, Xiaofei Xie, Sen Chen, Lingxiao Jiang, Xiaohong Li. Dissecting Global Search: A Simple Yet Effective Method to Boost Individual Discrimination Testing and Repair. In Proceedings of the 47th IEEE/ACM International Conference on Software Engineering, 2025, Accepted. (ICSE'25). [**CCF-A**]
5. Yuekun Wang, **Lili Quan**, Xiaofei Xie, Junjie Wang, Jianjun Chen, SPEC2CODE: Mapping Protocol Specification to Function-Level Code Implementation, In 2020 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025)[**CCF-A: Corresponding Author**]
6. **Lili Quan**, Jin Wen, Qiang Hu, Maxime Cordy, Yuheng Huang, Lei Ma, Xiaohong Li, Evaluation and Improvement of Test Selection for Large Language Models, In Journal of Software: Evolution and Process. (2025)[**CCF-B**]

## Awards

2015.10 **Merit Student**
2016.10 **National Encourage scholarship**
2019.05 **Second Prize of Information Security Network Attack and Defense Competition**
2021.09 **Doctoral Special Fellowship of Tianjin University**

## Academic Service

- PC member of the 1st International Workshop on Large Language Model Supply Chain Analysis (LLMSC 2025)
- PC member of the International World Wide Web Conference 2025 (WWW 2025)
- Junior PC member of Conference on Mining Software Repositories (MSR 2026)
- PC member of FSE 2026, Track of Ideas, Visions and Reflections (FSE2026-IVR)
- Reviewer of Journal of Software: Evolution and Process.

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